

1. Y_i take values in $\{3, 4\}$

$$f_{Y_i}(3) = c(\theta)\theta$$

$$f_{Y_i}(4) = c(\theta)\theta^3$$

$$1 = f_{Y_i}(3) + f_{Y_i}(4) = c(\theta)\theta + c(\theta)\theta^3 \\ = c(\theta)(\theta + \theta^3)$$

$\Rightarrow$

$$c(\theta) = \frac{1}{\theta + \theta^3} \quad \text{and} \quad f_Y(3) = \frac{1}{1 + \theta^2}, \quad f_Y(4) = \frac{\theta^2}{1 + \theta^2}$$

Data : 4, 4, 3

$$\text{Likelihood } L(\theta) = f_Y(4) \cdot f_Y(4) \cdot f_Y(3)$$

$$= \left(\frac{\theta^2}{1 + \theta^2}\right)^2 \cdot \frac{1}{1 + \theta^2}$$

$$= \frac{\theta^4}{(1 + \theta^2)^3}$$

$$\text{Loglikelihood } l(\theta) = \log L(\theta)$$

$$= 4 \log \theta - 3 \log(1 + \theta^2)$$

$$l'(\theta) = \frac{4}{\theta} - \frac{6\theta}{1 + \theta^2} = 0$$

$\Leftrightarrow$

$$4 + 4\theta^2 - 6\theta^2 = 4 - 2\theta^2 = 0$$

$\Leftrightarrow$

$$2 - \theta^2 = 0 \Leftrightarrow \theta = \pm\sqrt{2} \quad \text{and only } \sqrt{2}$$

possible since $(\mathcal{H}) = \mathbb{R}_+$

Sign study of $l'(\theta)$ shows that $\theta = \sqrt{2}$ is a maximum. Thus $\hat{\theta} = \sqrt{2}$

(1)

Solutions

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inference theory

MA1A02/

AMAD04

2. Data

$x_1, \dots, x_7$ obs of X_i with $E(X_i) = \theta$

$y_1, \dots, y_4$ obs of Y_i with $E(Y_i) = 3\theta$

sum of squares

$$Q(\theta) = \sum_{i=1}^7 (x_i - \theta)^2 + \sum_{i=1}^4 (y_i - 3\theta)^2$$

$$Q'(\theta) = -2 \sum_{i=1}^7 (x_i - \theta) - 2 \sum_{i=1}^4 3 (y_i - 3\theta)$$

$$= 86\theta - 2\sum x_i - 6\sum y_i = 0$$

$\Rightarrow$

$$\hat{\theta} = \frac{2\sum x_i + 6\sum y_i}{86} = 1.4991$$

This is a minimum value since

$Q(\theta)$ is convex

(Alt. $Z_i = Y_i/3$ has $E(Z_i) = \theta$,

and $\hat{\theta}$ is the average of

$$\{\tilde{x}_1, \dots, \tilde{x}_7\} = \{x_1, \dots, x_7, z_1, \dots, z_4\}$$

since it minimizes

$$Q(\theta) = \sum_{i=1}^{11} (\tilde{x}_i - \theta)^2$$

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3.

$$U \in \text{Exp}(1/\theta^2)$$

$$\Rightarrow$$

$$\theta^2 U \in \text{Exp}(1) \quad \text{since}$$

$$P(\theta^2 U \leq v) = P(U \leq \frac{v}{\theta^2}) = F_U(v/\theta^2)$$

$$= 1 - e^{-\theta^2(v/\theta^2)} = 1 - e^{-v} \quad (0)$$

when $v/\theta^2 \geq 0$ i.e. when $v \geq 0$

First we find c so that

$$0.95 = P(\theta^2 U \leq c) \quad (1)$$

when we have found that c , (1) gives

$$\begin{aligned} 0.95 &= P(\theta^2 \leq \frac{c}{U}) \\ &= P(-\sqrt{\frac{c}{U}} \leq \theta \leq \sqrt{\frac{c}{U}}) \end{aligned}$$

and thus

$$I_\theta = (-\sqrt{\frac{c}{U}}, \sqrt{\frac{c}{U}})$$

will be a two-sided CI with confidence grade 0.95.

But c in (1) is found by (0)

$$0.95 = 1 - e^{-c}$$

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$$\Leftrightarrow e^{-c} = 0.05$$

$$\Leftrightarrow c = -\log(0.05)$$

Thus

$$\begin{aligned} \bar{I}_\theta &= \left(-\sqrt{\frac{-\log(0.05)}{3.5}}, \sqrt{\frac{-\log(0.05)}{3.5}} \right) \\ &= (-0.9255, 0.9255) . \end{aligned}$$

~~QED~~

$$4. \quad U = \bar{X}_1 + \dots + \bar{X}_n \in \text{Bin}(10, \theta)$$

$u = 4$ observation of U

$$H_0: \theta \geq 4/5$$

$$9) \quad H_1: \theta < 4/5$$

Reject H_0 for small values of U (since $\hat{\theta} = 4/10$ is the mle, var and plug-in estimator). The p-value method gives

$$p = \sup_{\theta \geq 4/5} P_{\theta}(U \leq 4) = P_{4/5}(U \leq 4)$$

$$= \left(\text{table 6, } n=10, p=0.2 \text{ with } U=10-V, V \in \text{Bin}(10, 0.2) \text{ if } U \in \text{Bin}(10, 0.8) \right)$$

$$= \cancel{P_{4/5}} P_{0.2}(10-V \leq 4) =$$

$\nwarrow$ note $\uparrow$ Bin(10, 0.2)

$$= P_{0.2}(V \geq 6) = 1 - P_{0.2}(V \leq 5) = 1 - F_V(5)$$

$$= 1 - 0.99368 = 0.0064$$

i.e. can reject e.g. on level 0.01.

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b) $n=1$ obs of $U \in \text{Bin}(10, \theta)$

$$H_0: \theta \geq \theta_0$$

$$H_1: \theta < \theta_0$$

Want to have

$$0.05 \geq \sup_{\theta \geq \theta_0} P_{\theta}(U \leq 1) = P_{\theta_0}(U \leq 1)$$

$$= P_{\theta_0}(U=0) + P_{\theta_0}(U=1)$$

$$= \binom{10}{0} \theta_0^0 (1-\theta_0)^{10-0} + \binom{10}{1} \theta_0^1 (1-\theta_0)^{10-1}$$

$$= (1-\theta_0)^{10} + 10 \cdot \theta_0 \cdot (1-\theta_0)^9$$

Since RHS is decreasing in θ , ~~it is~~ the RHS is largest at the left endpoint of the interval $(a, 1)$ for which/on which the above inequality is true. Thus we should solve for a in

$$0.05 = (1-a)^{10} + 10 \cdot a \cdot (1-a)^9$$

Using calculator and e.g. interval halving (or similar) we get $a \approx 0.39$

$$c) \quad P_{\theta}(U=1) = \binom{10}{1} \theta^1 (1-\theta)^9 =: \eta(\theta) \quad (2)$$

We want to estimate the parameter η , for which we have the relation $\eta(\theta)$ as in (2).

The plug-in estimator of η is then

$$\begin{aligned} \eta\left(\frac{1}{10}\right) &= \eta\left(\frac{4}{10}\right) \\ &= \binom{10}{1} \frac{4}{10} \left(1 - \frac{4}{10}\right)^9 \\ &\approx 0.0403 \end{aligned}$$

5. $x = 3$ observation of $\bar{X} \in \text{Bin}(50, \theta)$

a) Exact test on level 0.01 of

$$H_0: \theta \geq 0.1$$

$$H_1: \theta < 0.1$$

The p-value method gives

$$p = \sup_{\theta \geq 0.1} P_{\theta}(\bar{X} \leq 3) = P_{0.1}(\bar{X} \leq 3)$$

$$= P_{\bar{X}}(0) + P_{\bar{X}}(1) + P_{\bar{X}}(2) + P_{\bar{X}}(3)$$

$$= \binom{50}{0} 0.1^0 0.9^{50} + \dots + \binom{50}{3} 0.1^3 0.9^{47}$$

$$= P_{\bar{X}}(0) + \underbrace{0.0286}_{> 0.01} + P_{\bar{X}}(2) + P_{\bar{X}}(3)$$

$$> 0.01$$

so can not be rejected on level 0.01

b) May assume $n=50$ large enough to use Gaussian approximation, and thus by the CLT

$$\bar{X} \in N(50\theta, 50\theta \cdot (1-\theta))$$

so that

$$\frac{\bar{X} - 50\theta}{\sqrt{50\theta(1-\theta)}} \stackrel{d}{\sim} N(0,1)$$

Next by the LLN

$$\frac{\bar{X}}{n} \xrightarrow{P} \theta$$

and thus by Slutsky's theorem, we may ~~be~~ approximate

$$\frac{\bar{X} - 50\theta}{\underbrace{\sqrt{50 \cdot \frac{\bar{X}}{50} (1 - \frac{\bar{X}}{50})}}_{d(\bar{X})}} \stackrel{d}{\sim} N(0,1)$$

Therefore

$$0.95 \subseteq P\left(\frac{\bar{X} - 50\theta}{d(\bar{X})} \leq \lambda_{0.05}\right)$$

$$= P\left(\frac{\bar{X}}{50} - \frac{1}{50} d(\bar{X}) \lambda_{0.05} \leq \theta\right)$$

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The lower bounded CI, with approx. grade 0.95, is

$$I_{\theta} = \left(\frac{x}{50} - \frac{1}{50} d(x) \lambda_{0.05}, \infty \right)$$

Use $x=3$, $d(x) = \sqrt{50 \cdot \frac{3}{50} \left(1 - \frac{3}{50}\right)} \approx 1.6793$, $\lambda_{0.05} = 1.6449$.

c) If $\theta = 0.1$ we can use Gaussian approximation to get

$$T = \frac{\bar{X} - 0.1 \cdot 50}{\sqrt{50 \cdot 0.1(1-0.1)}} \stackrel{d}{\sim} N(0,1)$$

Thus a test for H_0 is

$$T < -\lambda_{0.01} \Rightarrow \text{reject } H_0 \quad (*)$$

and it has approx level 0.01.

Here $\lambda_{0.01} = 2.3263$ (from table 2)

The test gives

$$t = \frac{3 - 0.1 \cdot 50}{\sqrt{50 \cdot 0.1 \cdot 0.9}} = -0.9428 \not< -\lambda_{0.01}$$

and we do not reject H_0 .

The power function for the test (*) is

$$\pi(\theta) = P_{\theta}(T < -\lambda_{0.01})$$

$$= P_{\theta}\left(\frac{\bar{X} - 50 \cdot 0.1}{\sqrt{50 \cdot 0.1 \cdot 0.9}} < -\lambda_{0.01}\right)$$

$$= P_{\theta}\left(\frac{\bar{X} - 50 \cdot 0.1}{\sqrt{50 \cdot \theta(1-\theta)}} < -\lambda_{0.01} \sqrt{\frac{0.1 \cdot 0.9}{\theta(1-\theta)}}\right)$$

$$= P_{\theta} \left(\underbrace{\frac{\bar{X} - 50 \cdot \theta}{\sqrt{50 \cdot \theta (1 - \theta)}}}_{\in N(0,1)} < \frac{(0.1 - \theta) 50}{\sqrt{50 \cdot \theta (1 - \theta)}} - \lambda_{0.01} \sqrt{\frac{0.1 \cdot 0.9}{\theta (1 - \theta)}} \right)$$

$$= \phi \left(\frac{(0.1 - \theta) 50}{\sqrt{50 \cdot \theta (1 - \theta)}} - 2.3263 \sqrt{\frac{0.1 \cdot 0.9}{\theta (1 - \theta)}} \right)$$

$$\pi(0.05) = \phi \left(\frac{(0.1 - 0.05) 50}{\sqrt{50 \cdot 0.05 (1 - 0.05)}} - 2.3263 \sqrt{\frac{0.1 \cdot 0.9}{0.05 (1 - 0.05)}} \right)$$

$$= \phi(-1.5799) = 1 - \phi(1.5799)$$

$$= 1 - 0.9429 = \underline{0.0571}$$

6. U continuous random variable, d.f. F

$$\theta = E(U^3)$$

$$\mathcal{H} = \mathbb{R}_+$$

$u_1, \dots, u_{20}$ i.i.d. observations

$$E(U^3) = \int u^3 dF(u) = T(F)$$

a) The plug-in estimator of θ is

$$\hat{\theta}_n = T(F_n) = \int u^3 dF_n(u) = \frac{1}{n} \sum_{i=1}^n u_i^3$$

so that

$$\hat{\theta}_{20} = \frac{1}{20} \sum_{i=1}^{20} u_i^3 =$$

$$\begin{aligned} b) \quad E(\hat{\theta}_n) &= E\left(\frac{1}{n} \sum_{i=1}^n U_i^3\right) \\ &= \frac{1}{n} \sum_{i=1}^n E(U_i^3) = \frac{n\theta}{n} = \theta \end{aligned}$$

$$\begin{aligned} c) \quad E((\hat{\theta}_n - \theta)^2) &= E((\hat{\theta}_n - E(\hat{\theta}_n))^2) \\ &\quad \uparrow \\ &\quad \hat{\theta}_n \text{ unbiased} \\ &= \text{Var}(\hat{\theta}_n) = \text{Var}\left(\frac{1}{n} \sum_{i=1}^n U_i^3\right) \\ &= \frac{\text{Var}(U_1^3)}{n} = \frac{E(U_1^6) - (E(U_1^3))^2}{n} \end{aligned}$$

$\rightarrow 0$

; $n \rightarrow \infty$

Thus

$$\sigma_n^1 \xrightarrow{\sigma^2} \sigma ; n \rightarrow \infty$$